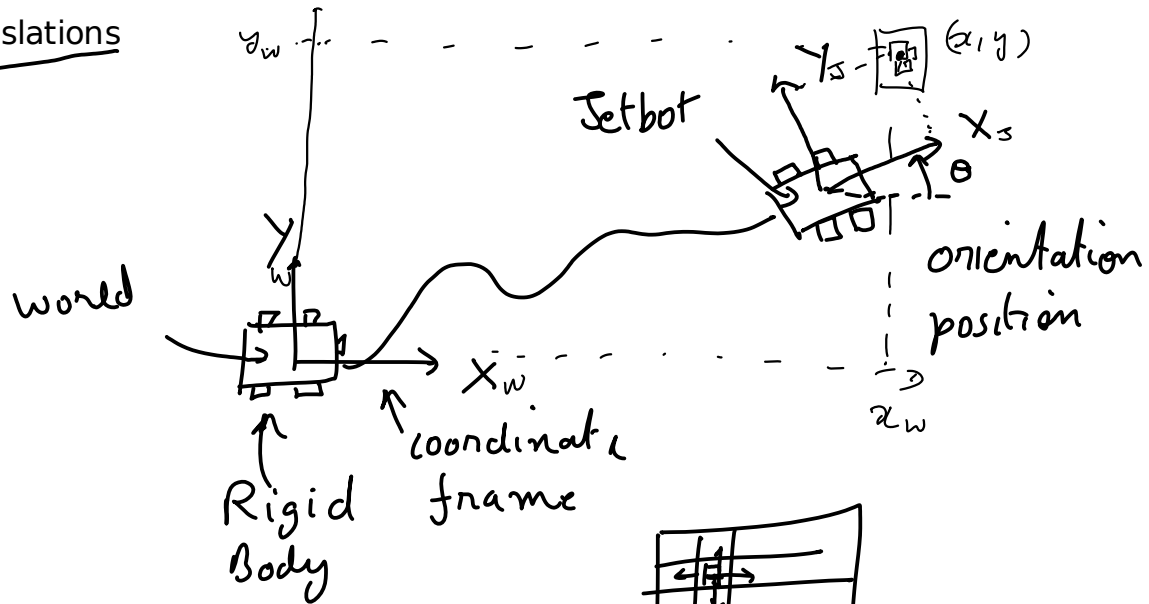
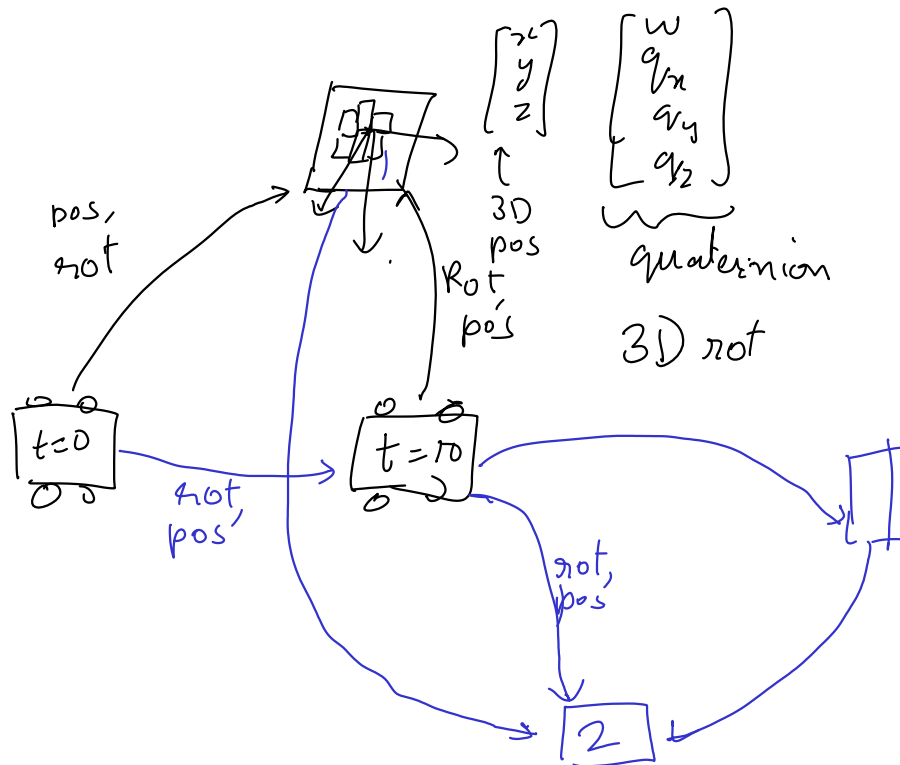


Rotations and translations



$$\begin{bmatrix} x_w \\ y_w \end{bmatrix} \xleftarrow[\text{transformation}]{\text{coordinate}} \begin{bmatrix} x_s \\ y_s \end{bmatrix}$$

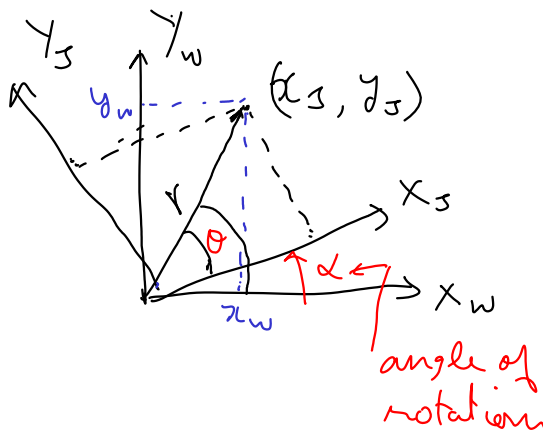


2D Rotation only

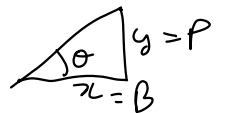
$$r = \sqrt{x_s^2 + y_s^2}$$

$$\theta = \text{arctan2}(y_s, x_s)$$

atan2



$$\tan^{-1}\left(\frac{y}{x}\right)$$



$$\begin{bmatrix} x_w \\ y_w \end{bmatrix} = \begin{bmatrix} r \cos(\theta + \alpha) \\ r \sin(\theta + \alpha) \end{bmatrix} = \begin{bmatrix} \underbrace{r \cos \theta}_{x_J} \cos \alpha - \underbrace{r \sin \theta}_{y_J} \sin \alpha \\ \underbrace{r \sin \theta}_{y_J} \cos \alpha + \underbrace{r \cos \theta}_{x_J} \sin \alpha \end{bmatrix}$$

$$x_J = r \cos \theta, \quad y_J = r \sin \theta$$

$$\underbrace{\begin{bmatrix} x_w \\ y_w \end{bmatrix}}_{2 \times 1} = \begin{bmatrix} x_J \cos \alpha - y_J \sin \alpha \\ y_J \cos \alpha + x_J \sin \alpha \end{bmatrix}$$

$$\underbrace{\underline{p}_w}_{2 \times 1} = \underbrace{\begin{bmatrix} \cos \alpha & -\sin \alpha \\ \sin \alpha & \cos \alpha \end{bmatrix}}_{\substack{\text{Rotation matrix} \\ 2 \times 2}} \underbrace{\begin{bmatrix} x_J \\ y_J \end{bmatrix}}_{\substack{\underline{p}_J \\ 2 \times 1}}$$

Dest.

$$\boxed{\underline{p}_w = {}^w R_J \underline{p}_J}$$

source

① Rotation of coordinate frame is a Linear transformation

② Rotation matrix depends only on α (the angle of rotation)

$$\boxed{{}^w R_J = \begin{bmatrix} \cos \alpha & -\sin \alpha \\ \sin \alpha & \cos \alpha \end{bmatrix}} \quad \text{③}$$

Properties of Rotation matrix

① Degrees of freedom in 2D Rotation = 1

② Determinant of Rotation matrix

$$\begin{vmatrix} \cos \alpha & -\sin \alpha \\ \sin \alpha & \cos \alpha \end{vmatrix} = \cos^2 \alpha + \sin^2 \alpha = 1 \quad \bigg| \quad \det(R) = 1$$

③

$$[{}^w R_J]^T [{}^w R_J]$$

$$= \begin{bmatrix} \cos \alpha & -\sin \alpha \\ \sin \alpha & \cos \alpha \end{bmatrix}^T \begin{bmatrix} \cos \alpha & -\sin \alpha \\ \sin \alpha & \cos \alpha \end{bmatrix}$$

$$= \begin{bmatrix} \cos^2 \alpha + \sin^2 \alpha & -\cancel{\cos \alpha \sin \alpha} + \cancel{\sin \alpha \cos \alpha} \\ -\cancel{\sin \alpha \cos \alpha} + \cancel{\cos \alpha \sin \alpha} & \cos^2 \alpha + \sin^2 \alpha \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I_{2 \times 2}$$

Rotation matrices are orthonormal

$$A^{-1} A = I$$

$${}^w R_J^{-1} = {}^w R_J^T$$

$$\underline{p}_w = {}^w R_J \underline{p}_J \xrightarrow[\text{multiply by } {}^w R_J^{-1}]{\text{Left}} \boxed{{}^w R_J^{-1} \underline{p}_w = \underline{p}_J}$$

$$\boxed{\underline{p}_J = {}^J R_w \underline{p}_w}$$

↑
world

$${}^J R_w = {}^w R_J^{-1} = {}^w R_J^T$$

$${}^w R_J = \begin{bmatrix} \cos \alpha & -\sin \alpha \\ \sin \alpha & \cos \alpha \end{bmatrix}, \quad {}^J R_w = \begin{bmatrix} \cos \alpha & \sin \alpha \\ -\sin \alpha & \cos \alpha \end{bmatrix}$$

(4) Orthonormal matrices have column vectors that are

a) Unit vectors

b) Perpendicular to each other